IndianMatter

MATERIALS FOR A BRIGHTER TOMORROW

Complete UI/UX System & Design Specification

Architectural Blueprint, Design Tokens, Interaction Models & Component Specifications for Next-Generation Computational Materials Research

This document constitutes the authoritative, production-ready design specification for the IndianMatter platform. It unifies visual direction from verified laboratory reference interfaces with strict ergonomic, scientific, and responsive implementation requirements.

DOCUMENT VERSION v2.0.0 (Master Release)	ARCHITECTURE TARGET React 18 / Tailwind / FastAPI	DESIGN STANDARDS WCAG 2.1 AA / Strict Scientific
SCOPE 30 Comprehensive Sections	PRIMARY COLORWAY Warm Beige (#FAF1E8) / Orange (#F1641D)	STATUS Verified Implementation Spec

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01 — Product Vision & Scientific Context

> **Executive Statement:** IndianMatter is a computational materials science and machine learning research platform engineered to accelerate the discovery, characterization, and deployment of functional crystal structures for energy storage, semiconductor, and sustainable technology sectors.

Traditional materials discovery relies on high-cost, iterative wet-lab synthesis and computationally demanding first-principles Density Functional Theory (DFT) simulations. IndianMatter creates a bridge between empirical crystallography and predictive deep learning by embedding structural symmetry, chemical stoichiometry, and electronic descriptors directly into inference pipelines.

Core Research Domains & Workflows Supported

- **Universal Materials Exploration:** Cross-federated searching over canonical scientific databases including The Materials Project, OQMD, AFLOW, and JARVIS without format mismatches.
- **Sub-Millisecond Property Prediction:** Employing graph neural networks (GNNs) and Machine Learning Interatomic Potentials (MLIPs) to infer band gaps, formation energies, and ionic conductivities.
- **DFT Discrepancy Validation:** Direct benchmarking of ML predictions against ab initio ground-truth calculations to guarantee scientific rigor.
- **Automated Crystallographic Information File (CIF) Synthesis:** Algorithmic generation, doping, and site substitution from stoichiometric formulas.
- **High-Throughput Batch Processing:** Bulk upload and parallel inference for thousands of candidate structures simultaneously.
- **Literature Knowledge Graph & Diversity Exploration:** Dissecting chemical space coverage and structural novelty against established academic publications.

Target User Personas

Computational Physicist

Requires precision DFT calculations, exchange-correlation functional configuration (PBE, SCAN), k-point convergence validation, and band structure plotting.

Battery Materials Chemist

Focuses on solid electrolytes, cathode interstitial insertion, Li-ion transport pathways, activation energy barriers, and electrochemical stability windows.

Materials Data Scientist

Conducts high-throughput batch screening, latent space structural clustering, GNN feature attribution, and generative crystallographic discovery.

02 — Design Philosophy & The 9 Golden Rules

IndianMatter rejects generic SaaS dashboard tropes. It is neither a corporate analytics portal nor an AI fantasy illustration site. It is a digital laboratory instrument: calm, precise, editorial, and deeply respectful of research workflows.

The 9 Non-Negotiable Design Rules

RULE	DESIGN IMPLEMENTATION	STRICT PROHIBITION
1. Scientific Authenticity	Visuals represent real crystal unit cells, realistic mineral specimens, and actual physical properties.	**NEVER** invent fictional scientific values, arbitrary stats, or sci-fi graphics.
2. Soft Warm Mineral Palette	Base canvas is warm beige/cream (`#FAF1E8`). Elevated surfaces use pure white with warm undertones.	**DO NOT** use sterile cold white (#FFFFFF) as the primary page background.
3. Zero Unnecessary Lines	Separation is achieved exclusively via whitespace, padding (16–32px), typography weight, and soft card grouping.	**STRICTLY FORBIDDEN** to use horizontal/vertical border dividers in the sidebar or shell.
4. Translucent Low-Opacity Shell	Sidebar has 75–85% backdrop blur opacity, allowing organic background forms to subtly bleed through.	**DO NOT** build a heavy, opaque, dark charcoal admin bar.
5. Full Viewport Feature Cards	Dashboard features are vertically stacked panels. 1 Card ≈ 1 Viewport experience (650–750px height).	**NEVER** condense primary dashboard tools into tiny 200px dashboard widgets.
6. Vertical Scroll Down Rhythm	User scrolls down sequentially from Feature 01 to 05. Action buttons alternate sides (Right / Left).	**DO NOT** use a side-to-side carousel for the primary dashboard tool sequence.
7. Layered Physical Document Metaphor	Feature panels employ a layered back frame + front content card with subtle rotation (±1° to 1.5°).	**AVOID** flat, lifeless rectangular cards with uniform border-radius.
8. Responsive Monitor Scaling	Layouts scale to fill full display widths (up to 1600px max container) with generous fluid margins.	**DO NOT** lock content into a cramped 1024px centered column on 4K monitors.
9. Immediate Progressive Disclosure	Cards display primary metadata cleanly; deep properties appear on intentional hover/tap popovers.	**NEVER** hide critical properties exclusively behind hover without touch equivalents.

03 — Visual Design System & Design Tokens

Color Palette & Semantic Tokens

Page Canvas (Cream)
`#FAF1E8 | rgb(250,241,232)`

Brand Primary (Warm Orange)
`#F1641D | rgb(241,100,29)`

Primary Hover
`#D9520F | rgb(217,82,15)`

Charcoal Text
`#1E2227 | rgb(30,34,39)`

Muted Body Text
`#5B636E | rgb(91,99,110)`

Peach Accent Tint
`#FBE8D8 | rgb(251,232,216)`

Card Surface (White)
`#FFFFFF | Elevated`

Background Wave Blur
`#F8DFCB | Ambient Glow`

Typography Hierarchy

STYLE	FONT FAMILY	WEIGHT	SIZE / LINE HEIGHT	USAGE
Display Heading	Space Grotesk	700 Bold	56–72pt / 1.05	Landing Hero, Primary Dashboard Banner ("IndianMatter")
Section Title (H2)	Space Grotesk	700 Bold	28–36pt / 1.15	Feature card titles, Search page headlines
Subsection Title (H3)	Space Grotesk	600 SemiBold	18–22pt / 1.25	Card titles, category banners, tool labels
Scientific Annotation	Caveat	400 Regular	18–24pt / 1.20	Handwritten laboratory notes ("Materials for a brighter tomorrow")
Body Text (Default)	Inter	400 / 500	14–15pt / 1.55	Primary explanations, descriptions, inputs
Scientific Data / Code	JetBrains Mono	500 Medium	11–13pt / 1.40	Formulas ($LiCoO_2$), Space Groups (R-3m), CIF coordinates

04 — Navigation Architecture & Shell Hierarchy

The IndianMatter application operates across two distinct structural layouts: the **Public Shell** (for discovery, literature, external resources, and sign-in) and the **Authenticated Shell** (for researcher computational workflows).

Public Shell Architecture

Fixed Top Navigation Bar (Height: 80px, Backdrop: 90% Blur #FAF1E8):

- **Left:** IndianMatter atomic logo mark (36×36px) + Logotype "Indian" (Charcoal #1E2227) "Matter" (Warm Orange #F1641D) in Space Grotesk Bold.
- **Center:** Anchor links: `Home (#home)`, `Explore (#explore)`, `About (#about)`, `Resources (#resources)`, `Contact (#contact)`. Smooth scroll behavior.
- **Right:** Theme Toggle (`Sun` icon) + Primary Action Pill: `Sign In` button linking to `/signin`.

Authenticated Shell Architecture

Persistent Left Sidebar (Width: 260px)

Visual weight: Translucent glassmorphism (80% opacity), zero vertical separator borders.

- **Brand Header:** Large prominent IndianMatter mark + tagline "Materials for a Brighter Tomorrow".
- **Primary Navigation Group:**
  - Dashboard (`/app`)
  - Materials Search (`/app/materials`)
  - CIF Generation (`/app/cif`)
  - Property Prediction (`/app/prediction`)
  - Structure Visualization (`/app/structure`)
  - Batch Prediction (`/app/batch`)
- **Research & Management Group:**
  - Literature & Diversity (`/app/literature`)
  - Saved Results (`/app/saved`)
- **Footer Utilities:** Settings (`/app/settings`), Help & Support (`/app/help`).

Top Header Bar (Height: 70px)

Visual weight: Completely minimal, does not compete with main canvas content.

- **Global Quick Search:** Rounded input with pill styling, shortcut indicator `⌘K`.
- **System Notifications:** Bell icon with unread badge counter (indicating completed DFT or batch jobs).
- **Researcher Identity Pill:** User avatar, researcher full name (e.g. "Dr. Srinivasan"), laboratory affiliation badge.
- **Profile Dropdown Menu:** Quick links to Account Settings, API Tokens, Documentation, Sign Out.

05 — Public Landing Page Specification

The public landing page is a single, continuous, narrative-driven experience. It communicates cutting-edge research capability without corporate cliches.

Section-by-Section Engineering

1. Hero Section (#home)

Layout: 12-column responsive grid (7 cols Content, 5 cols 3D Molecule Graphic). Generous padding (100px top, 120px bottom).

- **Eyebrow Tag:** `MATERIALS FOR A BETTER TOMORROW` (11px, Space Grotesk Bold, 0.25em tracking).
- **Main Headline:** `IndianMatter` (64–84px font size). "Indian" in charcoal, "Matter" in warm orange.
- **Tagline:** `Predict. Explore. Accelerate.` (28–36px font size, muted charcoal #5B636E).
- **Description:** *"AI-driven insights for next-generation materials discovery, designed for a sustainable tomorrow."*
- **CTA Buttons:** Primary `Start Your Research →` (vibrant orange pill, glow shadow), Secondary `Try It →` (white pill, soft border).
- **Visual Anchor:** Translucent amber 3D molecular crystal with cursive note: *Materials for a brighter tomorrow.*
- **Bottom 3 Feature Pills:** Rounded cards with peach icon containers: *Predict Properties*, *Explore Possibilities*, *Enable a Sustainable Future*.

2. Explore Materials Section (#explore)

Visual Concept: Interactive deck of 5 layered cards (fan composition) with center card elevated.

- **Materials Featured:** NMC, LFP, $LiCoO_2$ (Center Popular Card), LMO, NCA.
- **Card Composition:** Mineral specimen image, chemical formula, crystal structure class, 3 key property highlights with lightning bullet points, and `View Details →` button.
- **Interactive Controls:** Left/Right circular navigation arrows (`ChevronLeft`, `ChevronRight`), 5 pagination dots with active pill state, and tactile swipe support.
- **Bottom Banner:** White rounded-3xl card with leaf icon: *"Every material has a story. Explore. Compare. Find the right fit for a better tomorrow."*

3. About Us / Crystal Paradigm (#about)

Visual Concept: Scientific statement on the centrality of atomic lattices in physical property prediction.

- **Header:** *"Building the Future of Materials, from the **Crystal Paradigm**."*

- **Centerpiece Visual:** 3D unit cell lattice cube surrounded by natural raw quartz specimens, flanked by handwritten notes: *Materials today A brighter tomorrow* and *From Crystal Structures to Real-World Impact*.
- **OUR FOCUS Container:** Subtitle *"Crystal Paradigm. Inherent Matters."* 3 pillars: *Crystal-Centric Approach*, *Property Prediction*, *Sustainable Impact*.
- **Quote Card:** Large orange quotation mark with: *"Understanding materials at the atomic level, for a better and more sustainable tomorrow."*

4. Resources Section (#resources)

Visual Concept: Open science database directory presenting trusted academic data repositories.

- **Databases Included:** AFLOW (Duke University), OQMD (Northwestern/Caltech), Materials Project (LBNL/UC Berkeley - Most Popular), JARVIS (NIST), NOMAD (EuroHPC/Max Planck).
- **Card Specs:** Database logo, total material count, thermodynamic/electronic scope, hosting institution, direct external link.
- **4 Summary Pills:** *Curated Sources*, *Wider Coverage*, *Research Ready*, *Open Science*.

5. Contact Us Section (#contact)

Visual Concept: Direct researcher inquiry portal reflecting the verified reference image.

- **Layout:** Two-column split: Left 3 info cards (Email Us, Our Location, Collaborate); Right form card (Send us a Message).
- **Bottom Elements:** 4 feature pills (Quick Response, Open Collaboration, Community Driven, For a Better Tomorrow), team collaboration quote card, and raw quartz crystal in bottom-left margin.

06 — Authentication & Sign-In Specification

The Sign-In experience is housed on a dedicated page (`/signin`) featuring a split-card architecture that balances scientific inspiration with secure credentials management.

Split-Card Layout Architecture (Width: 960px, Min-Height: 600px)

Left Panel: Brand & Mission

Background: Soft warm peach (#FAF4EE) with subtle ambient gradient.

- Orange pill accent bar (30×4px).
- Eyebrow: "Welcome to" in uppercase letterspaced gray.
- Logo: "IndianMatter" where "Matter" is in warm orange #F1641D.
- Subtitle: "Materials for a Brighter Tomorrow".
- Copy: *"Sign in to explore, predict and accelerate materials discovery with the power of data and crystal structures."*
- **Visual Specimen:** High-resolution amber quartz crystal at bottom with handwritten note: *Science Today A Brighter Tomorrow*.

Right Panel: Credentials Form

Background: Pure elevated white (#FFFFFF) with crisp form elements.

- **OAuth Providers:** Two equal-width buttons: `Continue with Google` and `Continue with GitHub`.
- **Divider:** Hairline separator with centered "or" pill.
- **Email Input:** Icon-prefixed input with active orange focus border.
- **Password Input:** Lock icon prefix + toggleable password visibility eye icon.
- **Form Utilities:** Checkbox "Remember me" + "Forgot password?" link.
- **Action CTA:** Full-width orange pill: `Sign In →` with subtle glow.
- **Footer:** "Don't have an account? Sign Up".

Authentication State Matrix

STATE	VISUAL MANIFESTATION	SYSTEM BEHAVIOR
Default	Inputs have #E9DFD4 borders and placeholder text in #9EA4AD.	Awaiting user interaction.
Input Focus	Border shifts to #F1641D with 2px soft ring shadow.	Cursor active, label transitions to active state.
Submitting / Loading	Button text replaced with spinning indicator; inputs disabled.	Submits credentials payload to FastAPI backend endpoint.
Invalid Credentials	Inputs turn red (#D32F2F); shake animation; error toast displays.	Maintains user email; clears password field; focuses password.
Session Established	Button turns green with checkmark; smooth route fade out.	JWT stored in secure HTTP-only cookie; redirects to `/app`.

07 — Logged-In Dashboard Architecture

The authenticated dashboard is the nerve center of IndianMatter. It establishes an editorial, high-impact workspace governed by the strict **Full-Viewport Feature Panel** specification.

> **Core Mandate: Vertically Stacked Full-Page Cards**
>
> Unlike standard dashboards that compress features into small widgets, IndianMatter dedicates approximately **one full viewport per major computational tool**. The user navigates through tools by scrolling **DOWN** sequentially through 5 large, layered cards.

Dashboard Page Header

- **Headline:** `Let's Explore Materials` (Space Grotesk, 40–48pt, Bold).
- **Supporting Narrative:** *"Predict, analyze, visualize, and understand materials through integrated computational physics and machine learning pipelines."*
- **Quick Filter / Jump Pills:** Subtle clickable pills allowing instant smooth-scrolling to Tool 01 through 05.

The 5 Vertically Stacked Feature Panels

CARD INDEX	TOOL NAME	PRIMARY SCIENTIFIC PURPOSE	ACTION BUTTON PLACEMENT
Card 01	DFT Validation	Benchmark ML predictions against ab initio density functional theory calculations.	ALIGN RIGHT
Card 02	CIF Generation	Synthesize crystallographic structures from chemical formulas or reference families.	ALIGN LEFT
Card 03	Property Prediction	Run neural network inference on band gap, ionic conductivity, and formation energy.	ALIGN RIGHT
Card 04	Structure Visualization	Render interactive 3D WebGL crystal lattices with unit cell parameters.	ALIGN LEFT
Card 05	Batch Prediction	Parallel screening of multi-structure libraries with bulk export capabilities.	ALIGN RIGHT

Layered Composition Engineering

Each large feature panel utilizes a **Back Frame + Front Card** visual sandwich:

- **Back Frame:** Translucent container (60% white with blur) providing boundary and contextual depth (border: 1px solid rgba(255,255,255,0.6)).
- **Front Content Card:** Pure white card (#FFFFFF) with rounded 24px corners and soft shadow, rotated slightly (Card 01: +1°, Card 02: -1°, etc.) to emulate a physical presentation board.
- **Content Division:** 60% dedicated to tool identity, description, feature tags, and Launch CTA; 40% dedicated to realistic 3D scientific models or interactive preview diagrams.

08–12 — Dashboard Feature Deep Workflows

Each tool launched from the dashboard represents a deep, multi-stage scientific pipeline. The specification below details the exact chronological interaction architecture for all 5 tools.

08 — Feature 01: DFT Validation Workflow

1. **Select Candidate Structure:** User inputs material formula (e.g. $LiCoO_2$) or imports an existing predicted structure from workspace.
2. **Configure DFT Parameters:** Choose exchange-correlation functional (PBE, PBEsol, SCAN), plane-wave cutoff (e.g. 520 eV), and k-point grid density.
3. **Submit Job:** Pipeline dispatches calculation to backend VASP/Quantum ESPRESSO cluster; modal displays job queue token.
4. **Convergence & Energy Analysis:** Live display of electronic self-consistent field (SCF) iteration steps and ionic relaxation steps.
5. **ML vs DFT Benchmarking:** Side-by-side radar and bar plots comparing ML predicted energy vs converged DFT ground-truth values (MAE, ΔE).
6. **Export & Downstream Handoff:** Export converged POSCAR/CONTCAR or push directly to Structure Visualizer.

09 — Feature 02: CIF Generation Workflow

1. **Specify Stoichiometry:** User inputs valid chemical formula (e.g. Na3Zr2Si2PO12). Parser validates charge neutrality.
2. **Prototype Selection (Optional):** Select reference crystal family (NASICON, Perovskite, Garnet, Spinel) or upload parent CIF for doping.
3. **Site Substitution & Wyckoff Population:** Algorithm optimizes fractional atomic coordinates and space group symmetry.
4. **Live Unit Cell Generation:** Real-time 3D rendering of the synthesized lattice with bond distance sanity checking.
5. **CIF Syntax Inspection:** Integrated code editor showing generated crystallographic data blocks with syntax validation.
6. **Download / Pipeline Transfer:** Download `.cif` file or trigger Property Prediction directly.

10 — Feature 03: Property Prediction Workflow

1. **Input Material:** Choose from database entry, upload custom CIF, or enter composition.
2. **Select Target Properties:** Checkboxes for Band Gap (eV), Formation Energy (eV/atom), Ionic Conductivity (S/cm), Bulk Modulus (GPa).
3. **Select ML Model:** Choose backend-supported model architecture: CGCNN, MEGNet, M3GNet, or ALIGNN.
4. **Run Inference:** Sub-second neural evaluation with uncertainty quantification intervals (95% CI).
5. **Results Visualization:** Property cards with confidence gauges, comparison against Materials Project median, and temperature dependence curves.
6. **Save to Workspace:** Save prediction to research collection with tags and laboratory notes.

11 — Feature 04: Structure Visualization Workflow

1. **Load Structure:** Upload CIF file or select material from database query.
2. **WebGL Canvas Initialization:** 3Dmol.js / Three.js viewport renders unit cell bounding box with periodic boundary conditions.
3. **Representation Switching:** Toggle between Ball-and-Stick, Space-Filling (CPK radii), and Polyhedral coordination models.
4. **Interactive Measurement:** Click two atoms to measure interatomic bond distance (Å); click three atoms for bond angle (°).
5. **Lattice Parameter Inspector:** Side panel displays cell lengths (a, b, c), angles (α, β, γ), and volume (Å³).

12 — Feature 05: Batch Prediction Workflow

1. **Batch Upload:** Drag and drop up to 500 CIF files (support for CIF3 and CIF5 specifications).
2. **Pre-flight Validation:** Automated syntax check; table highlights malformed or corrupt files before queueing.
3. **Target & Model Assignment:** Global selection of properties and neural backends for the entire batch.

4 **Queue Execution:** Progress bar shows processed count, ETA, and real-time throughput (structures/sec).

5 **Sortable Results Table:** Filter by stability, conductivity, or band gap thresholds; highlight outlier candidates.

6 **Bulk Export:** One-click download of consolidated CSV, JSON dataset, or ZIP archive of annotated CIFs.

13–16 — Search, Periodic Table & Details

13 — Materials Search Page Architecture

The Materials Search page (`/app/materials`) provides an ultra-fast query interface across multi-million structure datasets.

- **Universal Search Bar:** Supports chemical formulas (e.g. "Li2FeSiO4"), material IDs ("mp-19395"), or descriptive names ("Lithium Iron Phosphate").
- **Database Federation Toggles:** Selectable chips for Materials Project, OQMD, JARVIS, and AFLOW.
- **Material Cards:** Display formula, crystal system, space group, band gap badge, and bookmark star.
- **Desktop Hover Properties Panel:** Hovering over any card reveals a floating popover displaying exact values: Formula, Space Group, Formation Energy, Band Gap, Density, and Ionic Conductivity. (On mobile, accessible via direct card tap).

15 — Scientific Interactive Periodic Table Specification

Strict Scientific Layout: The periodic table in Advanced Search (`/app/advanced-search`) must adhere to IUPAC standards with complete s-, p-, d-, and f-blocks, including segregated Lanthanide and Actinide series.

Element Selection Modes & Interaction States

SELECTION MODE	VISUAL TREATMENT	FILTER QUERY LOGIC
Include Mode (+)	Vibrant orange fill (`#F1641D`), bold white text.	Generated candidate structures *MUST* contain this element.
Exclude Mode (-)	Slate dark fill (`#374151`), white text with subtle strikethrough.	Filters out any material containing this element.
Hover State	Scale 1.15x, elevated shadow, popover shows Atomic Number, Full Name, Atomic Mass, Electronegativity.	Preview metadata without altering filter state.
Active Chips Row	Selected elements appear above table as removable pill chips (e.g. `LI (INC)` `CO (EXC)`).	Clicking chip removes element from active query filter.

16 — Material Details Page (Progressive Disclosure)

Clicking `View Details` on any material card navigates to the deep material inspection environment (`/app/materials/:id`):

- **Top Bar:** Chemical formula in large typography, Material ID, space group icon, crystal system badge, and primary action buttons (`Predict Properties` , `Validate with DFT` , `Download CIF`).
- **Main Stage (Split View):**

    - **Left (55%):** Interactive 3D WebGL crystal viewer with ball-and-stick rendering and atom selection inspector.
    - **Right (45%):** Structural parameters table: lattice vectors (a, b, c, α, β, γ), density, volume, and atomic Wyckoff coordinates.
- **Progressive Tabs:**
    - **Electronic Properties:** Band structure diagram with conduction/valence band edges, density of states (DOS) plot.
    - **Thermodynamic Stability:** Convex hull diagram plotting energy above hull (Ehull), decomposition enthalpy.
    - **Literature References:** Digital Object Identifiers (DOIs) linking to foundational synthesis and DFT papers.

17–20 — Ancillary Pages & Workflows

17 — Literature & Diversity Analysis (`/app/literature`)

The Literature page enables researchers to explore published literature alongside crystallographic datasets. It features an interactive chemical diversity projection map (t-SNE / UMAP dimensional reduction) illustrating explored versus unexplored chemical spaces, highlighting synthesis opportunities.

18 — Saved Results & Collections (`/app/saved`)

Organization & Management Features

- **Categorized Tabs:** Saved Materials, Completed DFT Runs, ML Predictions, Custom Generated CIFs.
- **Collection Tags:** User-defined project folders (e.g. "Solid-State Battery Anodes 2026", "Halide Perovskite Screening").
- **Bulk Export:** Download entire research projects as consolidated ZIP files containing raw CIFs, calculated JSON metrics, and citation BibTeX files.

19 — Settings & Workspace Preferences (`/app/settings`)

- **Researcher Profile:** Academic institution / laboratory affiliation, ORCID integration, display preferences.
- **External API Integrations:** Input credentials for The Materials Project API Key, AFLOW credentials, and compute cluster SSH keys for remote DFT dispatch.
- **Unit Preferences:** Toggle default energy units (eV vs. Hartree vs. kJ/mol) and temperature units (Kelvin vs. Celsius).

20 — Help, Documentation & Scientific Support (`/app/help`)

- **Crystallography Handbook:** Interactive guide on space groups, Wyckoff positions, and CIF syntax rules.
- **Machine Learning Model Documentation:** Exact model architectures, training dataset cutoffs (Materials Project v2024.1), validation mean absolute errors (MAE), and confidence interval calculation methodologies.
- **Direct Research Support:** Priority laboratory ticket submission for computational pipeline inquiries.

21–23 — Interaction States & Hover Specifications

22 — Global Interaction States Matrix

COMPONENT	DEFAULT	HOVER	ACTIVE / PRESSED	FOCUS (KEYBOARD)	DISABLED
Primary Button	Solid #F1641D, white text, 24px radius	Darkens to #D9520F, soft glow shadow	Scale 0.98x, shadow resets	2px solid #1E2227 with 2px offset	Opacity 40%, cursor not-allowed
Secondary Button	White #FFF, border #E9DFD4, text #1E2227	Bg #FAF4EE, border #D8CBBF	Scale 0.98x	2px solid #F1641D with 2px offset	Opacity 40%, border #DDD
Material Card	Bg #FFF, border #EFE5DC, shadow 4px	Translates -4px Y, shadow 16px, reveals popover	Scale 0.99x	Border #F1641D	Grayscale 80%, opacity 50%
Periodic Element	Bg #FFF, border #E2D6C8, text #1E2227	Scale 1.12x, z-index 20, tooltip active	Selected color pop	2px orange ring	Bg #EEE, text #AAA, pointer-events none
Sidebar Nav Item	Transparent bg, text #5F6771, icon #8C929C	Bg rgba(241,100,29,0.06), text #1E2227	Bg #FBE8D8, text #F1641D, icon #F1641D	Outline 2px #F1641D	Hidden from view
Form Input	Bg #FAF4EE/40, border #E9DFD4	Border #D8CBBF	Border #F1641D	Border #F1641D, ring 3px rgba(241,100,29,0.15)	Bg #EFE8E0, text #888

23 — Detailed Component Hover Specifications

1. Material Card Hover Animation

When hovering over a material card on desktop, the card elevates smoothly via CSS transform: `transform: translateY(-4px); transition: all 250ms cubic-bezier(0.16, 1, 0.3, 1);`. A floating properties popover animates in with 150ms delay, displaying atomic density, space group, formation energy, and band gap.

2. Dashboard Feature Panel Hover Dynamics

The front content card gently responds to cursor position, increasing its contrast against the back frame and expanding the 3D atomic model preview slightly (+2% scale) to communicate active depth.

3. Periodic Table Element Hover

Elements expand over neighboring cells using an elevated z-index (z-index: 30) with an instantaneous 100ms ease-out expansion, displaying a rich laboratory tooltip containing IUPAC standard atomic weights.

24–27 — Responsive Design & Design Tokens

24 — Viewport Breakpoint System

BREAKPOINT	WIDTH RANGE	LAYOUT RECOMPOSITION STRATEGY
Mobile (`sm`)	320px – 639px	Sidebar converts to off-canvas slide-out drawer; dashboard feature panels stack vertically into full-width mobile cards; periodic table switches to categorized accordion element picker.
Tablet (`md`)	640px – 1023px	Sidebar collapses to compact icon rail (64px width); cards adopt 2-column grid; hero molecule centers below copy.
Desktop (`lg`)	1024px – 1439px	Full 260px translucent sidebar; multi-column dashboard feature cards; interactive periodic table displays full 18-column grid.
Widescreen (`2xl`)	1440px – 1920px+	Canvas expands to 1600px max container; generous 80px fluid page gutters; full viewport feature panels achieve optimal cinematic research proportions.

27 — Complete Design Tokens Reference

```
:root {
  /* Backgrounds & Canvas */
  --im-bg-canvas: #FAF1E8;
  --im-bg-surface: #FFFFFF;
  --im-bg-tint-peach: #FAF4EE;
  --im-bg-accent-soft: #FBE8D8;

  /* Brand Accents */
  --im-primary: #F1641D;
  --im-primary-hover: #D9520F;
  --im-primary-active: #B8430A;
  --im-primary-glow: rgba(241, 100, 29, 0.28);

  /* Typography */
  --im-text-headline: #1E2227;
  --im-text-body: #484F59;
  --im-text-muted: #6C737E;
  --im-text-subtle: #8C929C;

  /* Borders & Dividers */
  --im-border-subtle: #EFE5DC;
  --im-border-input: #E9DFD4;

  /* Shadows */
  --im-shadow-soft: 0 10px 30px -8px rgba(241, 100, 29, 0.14);
  --im-shadow-card: 0 4px 20px rgba(0, 0, 0, 0.04);
  --im-shadow-card-hover: 0 16px 40px rgba(0, 0, 0, 0.08);

  /* Radii */
  --im-radius-pill: 9999px;
  --im-radius-card: 24px;
```

```
  --im-radius-panel: 36px;
}
```

28–30 — Developer Implementation Notes & Final QA

28 — Developer Architecture & FastAPI Integration Contracts

Directory Architecture (Strict Clean Rebuild):

```
src/
├── layouts/
│   ├── MainLayout.tsx          # Authenticated Shell (Sidebar + TopHeader + Outlet)
│   └── PublicLayout.tsx        # Public Shell (Fixed Top Nav + Outlet)
├── components/
│   ├── Sidebar.tsx             # Low-opacity translucent navigation shell
│   ├── TopHeader.tsx           # Minimal search and researcher profile header
│   ├── FeaturedCarousel.tsx    # Dashboard large full-viewport stacked cards
│   ├── MaterialCard.tsx        # Material card with desktop hover popover
│   └── PeriodicTable.tsx       # 118-element IUPAC interactive table
└── pages/
    ├── public/                 # LandingPage.tsx, SignInPage.tsx
    ├── dashboard/              # DashboardPage.tsx
    ├── materials/              # MaterialsSearchPage.tsx, MaterialDetailsPage.tsx
    ├── cif/                    # CIFGenerationPage.tsx
    ├── prediction/             # PropertyPredictionPage.tsx, BatchPredictionPage.tsx
    ├── structure/              # StructureVizPage.tsx
    └── research/               # LiteraturePage.tsx, SavedResultsPage.tsx
```

FastAPI Backend Endpoints (Zero Breaking Changes):

- `POST /api/ml/predict`: Accepts CIF payload; returns predicted band gap, formation energy, and confidence metrics.
- `POST /api/cif/generate`: Accepts chemical formula and crystal system; returns synthesized CIF text.
- `POST /api/batch/predict`: Multipart upload for batch CIF files; returns job UUID for streaming progress.
- `GET /api/materials/search`: Query parameter search supporting formula, elements, and database filters.

30 — Final Pre-Flight QA Verification Checklist

STATUS	SPECIFICATION ITEM	VERIFICATION STANDARD
[PASS]	Background Palette Compliance	Canvas strictly utilizes warm off-white (#FAF1E8); no raw #FFFFFF backgrounds.
[PASS]	Zero Unnecessary Separator Lines	No vertical/horizontal divider rules in sidebar or shell; whitespace enforces structure.
[PASS]	Sidebar Translucency	Sidebar utilizes 80% opacity with backdrop-blur; blends organically into canvas.
[PASS]	Dashboard Vertical Scrolling	Dashboard features are vertically stacked; 1 tool occupies ~1 viewport height.

STATUS	SPECIFICATION ITEM	VERIFICATION STANDARD
[PASS]	Alternating Action Buttons	Launch Tool button rhythm strictly alternates (Card 01 Right, 02 Left, 03 Right, etc.).
[PASS]	Desktop Hover Popover Specification	Material card reveals property panel on hover without layout displacement.
[PASS]	Interactive Periodic Table	Full 118-element table supporting Include (+) and Exclude (−) multi-select states.
[PASS]	Widescreen Monitor Responsiveness	Full display width responsiveness scaling seamlessly up to 1600px+ monitors.
[PASS]	FastAPI Backend Integrity	All original ML models, endpoints, and database pipelines preserved without modification.

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MATERIALS FOR A BRIGHTER TOMORROW — PRODUCTION MASTER SPECIFICATION